

Intermediate Phonology

Part 7: Prosodic phrase and intonation phrase

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Contents of the class

1. Prosodic hierarchy; Phonemes, segments, and distinctive features
2. Segmental allophonies
3. Syllables
4. Moras, weight and time slots
5. Feet and lexical stress
6. Prosodic words
7. **Prosodic phrases**
8. Tones and intonation

Prosodic hierarchy

Prosodic structure is crucial to understand phonology

Prosody

ι	Intonation phrase
φ	Prosodic Phrase
ω	Prosodic word
F	Foot
σ	Syllable
μ	Mora

Morphosyntax

Clause
Syntactic phrase
Grammatical word
Stress unit
–
–

ω, φ and ι : interface constituents

Introduction

Mapping between syntax and prosody takes place at the higher-level prosodic constituents: ideally, a syntactic phrase is mapped to a prosodic phrase (φ -phrase), and a clause is mapped to an intonation phrase (ι -phrase).

ι	Intonation phrase	Clause
φ	Prosodic Phrase	Syntactic phrase

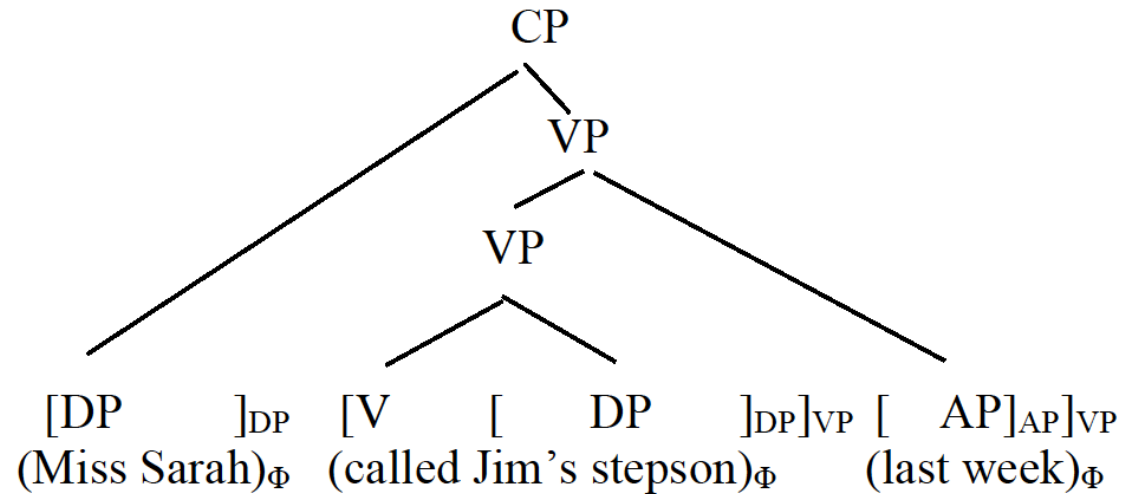
In an all new utterance, only syntax is active in the mapping to prosodic structure.

Information structure, i.e., focus, givenness and topic, can have an influence not only on the syntactic structure of sentences, but also on the prosody. For instance, the placement of “nuclear” accents can be determined by focus structure.

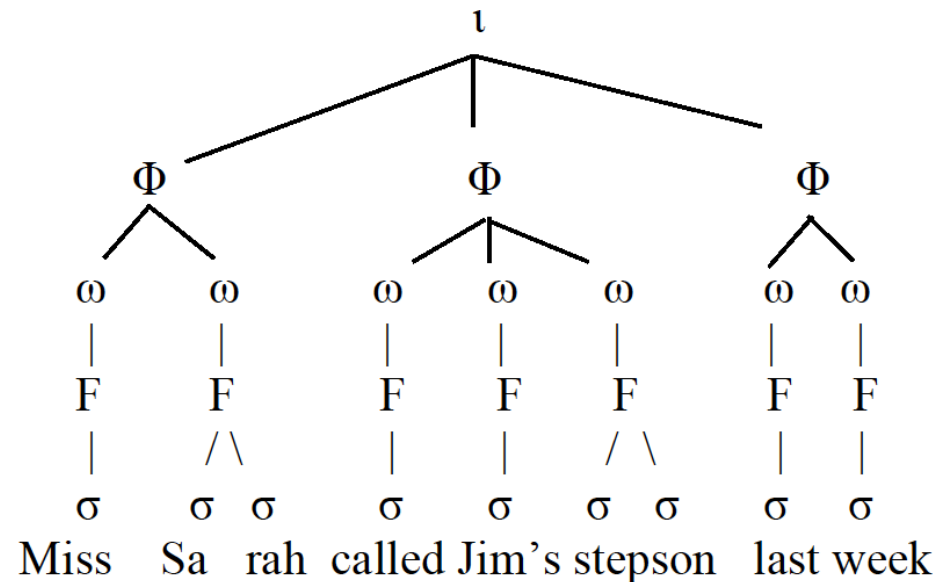
Prosodic phrasing in English

An example of mapping
between syntax and prosody
in an all-new sentence

Syntactic structure



Prosodic structure



Prosodic phrasing in English

Two main models of syntax-prosody interface:

1. **Alignment model** (there are equivalent constraints for the left side of syntactic phrases)

a. ALIGN-XP,R: ALIGN(XP, R; φ , R)

For each XP there is a φ such that the right edge of XP coincides with the right edge of φ .

b. ALIGN-XP,L: ALIGN(XP, L; φ , L)

For each XP there is a φ such that the left edge of XP coincides with the left edge of φ .

2. **Match model**

(i) MATCH Phrase

A phrase in syntactic representation must be matched in phonological representation by a constituent of type φ .

(ii) MATCH Clause

A clause in syntactic representation must be matched in phonological representation by a constituent of type ι .

Prosodic phrasing in English

In principle, both ALIGN and MATCH allow recursive structures. In practice, only MATCH allows recursivity.

In ALIGN, one edge of syntactic constituents, left **or** right, is aligned with a φ -phrase or an ι -phrase.

Predictions of ALIGNR

- a. Miss Martin) $_{\varphi}$ went to the market) $_{\varphi}$ with a basket full of eggs) $_{\varphi}$) $_{\iota}$
- b. Miss Sarah) $_{\varphi}$ called Jim's stepson) $_{\varphi}$ last week) $_{\varphi}$) $_{\iota}$

The other edge (the left one here) is also present because prosodic phrasing should be exhaustive.

In MATCH, both edges of syntactic constituents, left **and** right, are aligned with a φ -phrase or an ι -phrase at once.

Predictions of MATCH for English (more φ -phrases because of recursivity)

- a. ((Miss Martin) $_{\varphi}$ (went (to (the market) $_{\varphi}$) $_{\varphi}$) $_{\varphi}$ (with a basket (full (of eggs) $_{\varphi}$) $_{\varphi}$) $_{\varphi}$) $_{\iota}$
- b. ((Miss Sarah) $_{\varphi}$ (called (Jim's (stepson) $_{\varphi}$) $_{\varphi}$) $_{\varphi}$ (last week) $_{\varphi}$) $_{\iota}$

Selkirk, Elisabeth O. 2011. The syntax–phonology interface. In: Goldsmith, John A., J. Riggle & A. Yu (eds.). *The handbook of phonological theory 2*. Oxford: Blackwell. 435-484.

Prosodic phrasing in English

Faithfulness constraints: ALIGN constraints or MATCH constraints

Markedness constraints (sometimes called “Wellformedness constraints”) for higher-level prosodic constituents (φ -phrases and ι -phrases) account for Binarity (a φ -phrase consists of two ω -words), Exhaustivity (no level of the prosodic hierarchy is skipped), Layeredness (lower constituents do not dominate higher constituents), Culminativity, location of nuclear accent, ...

The wellformedness constraints can overwrite the effects of the interface constraints: Even though they are XP, pronouns are too weak to form their own φ -phrase and are cliticized to the adjacent φ -phrase, resulting in a mismatch between syntax and prosody (see also Elfner 2015 and colleagues for Irish clitics and further mismatches).

Elfner, Emily. 2015. Recursion in prosodic phrasing: Evidence from Connemara Irish. *Natural Language and Linguistic Theory*.

Prosodic phrasing in English

Kratzer & Selkirk (2020) propose a stratal derivation of the syntax-prosody interface

Three distinct levels of grammatical representation: (a) is the morphosyntactic output (MSO) representation of an all-new declarative sentence; (b) is the phonological input (PI) representation of the sentence resulting from the MATCH constraints; (c) is the phonological output (PO) representation, which is submitted to the wellformedness constraints and the phonetic interpretation.

Morpho-Syntactic Output Representation (MSO)

↓ Spellout (gives phonological expression to MSO)

Phonological Input Representation (PI)

↓ Phonology (determines optimal PO on basis of PI)

Phonological Output Representation (PO)

↓

Phonetic Interpretation

Kratzer, Angelika and Elisabeth Selkirk. 2020. Deconstructing information structure. *Glossa: a journal of general linguistics* 5(1): 113. 1–53.

Prosodic phrasing in English

Sarah mailed the caramels

- Morphosyntactic output representation (MSO):

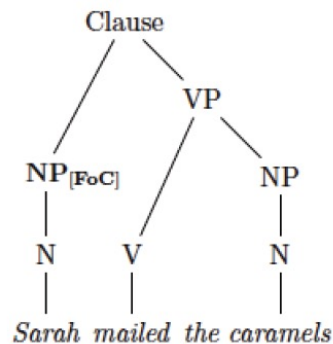
$$[[[Sarah]_N]_{NP} [[mail-ed]_V [the [[caramel-s]_N]_{NP}]_{DP}]_{VP}]_{Clause}$$
- Phonological input representation (PI):

$$((Sarah)\omega)\varphi((mailed)\omega(the((caramels)\omega)\varphi)\varphi)\iota$$
- Phonological output representation (PO):

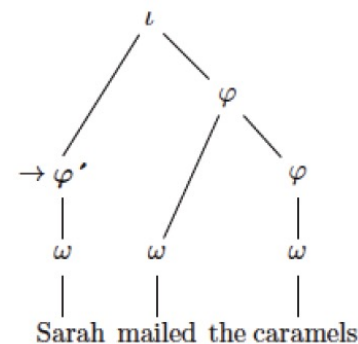
$$(((S\acute{a}r.ah)\omega')\varphi)((m\acute{a}iled)\omega \text{ the)\omega((c\acute{a}r.a.m\grave{e}ls)\omega')\varphi')\iota$$

H L ~H H L

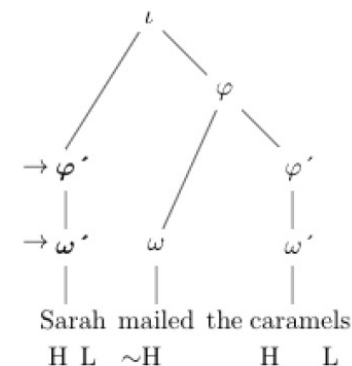
a. Morphosyntactic Output³⁹



b. Input phonological representation (PI)



c. Output phonological representation (PO)



Prosodic phrasing in English

The unmarked position of nuclear stress is on the head of the last φ -phrase, usually an argument or an adjunct:

$((\text{Sarah})_{\varphi} (\text{mailed (the CARMELS)}_{\varphi})_{\varphi})_{\iota}$

$((\text{Miss Martin})_{\varphi} (\text{went (to (the market)}_{\varphi})_{\varphi})_{\varphi} (\text{with a basket (full (of EGGS)}_{\varphi})_{\varphi})_{\varphi})_{\iota}$

In an argument predicate sequence, the argument gets the nuclear accent, even if it is not the last element of the sentence.

- a. What did Mary do?/Whom did Mary praise?
- b. She praised her BROTHER
- c. She had her HAIR done.
- d. She wants the WALL painted.
- e. Which BOOKS did you read?

Prosodic phrasing in English

The main effect of information structure is to add a nuclear accent on the focused constituent and to deaccent given constituents.

(1) a. What did John's mother do?

b. She [[PRAISED]_F him]_F

(2) a. I know that John drove Mary's red convertible. But what did Bill drive?

b. He drove [her BLUE]_F convertible_G]_F.

(3) a. Why did John buy bananas?

b. Because [[SARAH]_F [bought bananas]_G]_F

(4) a. Sarah said the American president drinks. What did Lisa say about the French president?

b. She said [[[the French president drinks]_G [TOO]_F]_F

Schwarzschild, Roger (1999) GIVENness, AvoidF and Other Constraints on the Placement of Accent. *Natural Language Semantics* 7. 141-177.

Féry, Caroline & Vieri Samek-Lodovici (2006) Focus projection and prosodic prominence in nested foci. *Language* 82.1. 131-150.

Prosodic phrasing in English

Kratzer & Selkirk do not consider all-new sentences, such as (1) as focused. Only a narrow focus is a focus (2) and (3).

- (1) Me: Did anybody eat the clementines? I can't find them in the pantry.
You: (I think) Paula might [have eaten the clementines]_G.
- (2) Me: Sarah mailed the caramels.
You: (No), [Eliza]_{FoC} [mailed the caramels]_G.
Aunt: (Yes, and) [Ewan]_{FoC} [mailed]_G [the chocolates]_{FoC}
- (3) Me: Jane's lost her keys and is really upset.
You: It was her phone that Jane lost.

Summary for English

The mapping can be symmetrical on both sides (MATCH), or it can take place first on one side only (ALIGN).

Boundaries between φ -phrases are “asymmetric”. They are often much more audible on one side than on the other. Compare the English sentences (1) and (2):

(1) (Anna thinks φ (her cat is a lion φ))_t

(2) ((That the temperature is so high) φ (is a problem for all of us) φ))_t

The right boundary is more prominent in (2) than the left one in (1). The location of nuclear stress correlates with the prominent boundary.

MATCH disregards this difference, ALIGN emphasizes it.

Ito & Mester (2019) and Myrberg & Ishihara (2023) propose extensions and changes to Kratzer & Selkirk (2020) to account for the “asymmetry problem” in Japanese.

The discussion around Align and Match is unresolved so far.

Ito, Junko & Armin Mester. 2019. Match Theory and prosodic well-formedness constraints. In Hongming Zhang & Youyong Quian (eds.). *Prosodic Studies: Challenges and prospects*. 252–274. London: Routledge.

Myrberg, Sara & Shinichiro Ishihara. 2023. Match Theory and the Asymmetry Problem: An example from Stockholm Swedish. In *Languages* 6: 65.